Geometry Review: Unit 5 — Right Triangles & Trigonometry

Station 1 of 4: The Pythagorean Theorem

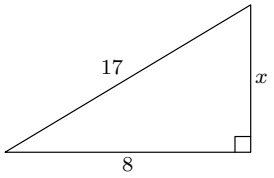
Name: _____ Date: _____

Use $a^2 + b^2 = c^2$. Give exact answers in simplest radical form, and round decimals to the nearest tenth when asked. Diagrams are not drawn to scale.

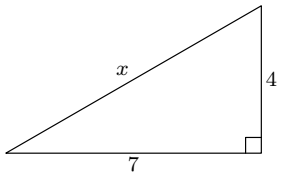
1. The legs of a right triangle are 6 and 8. Find the hypotenuse.

2. The legs of a right triangle are 5 and 12. Find the hypotenuse.

3. The hypotenuse of a right triangle is 17 and one leg is 8. Find the other leg.



4. The legs of a right triangle are 4 and 7. Find the hypotenuse. Give the exact answer and a decimal to the nearest tenth.

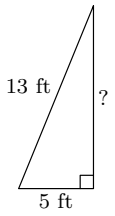


5. The hypotenuse of a right triangle is 9 and one leg is 5. Find the other leg in simplest radical form and as a decimal to the nearest tenth.

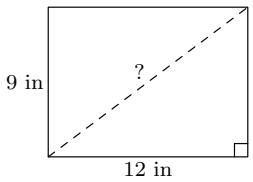
6. Is $9, 12, 15$ a Pythagorean triple? Explain.

7. A triangle has sides $7, 8, 12$. Is it acute, right, or obtuse?

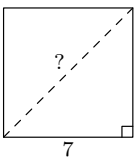
8. A 13-ft ladder leans against a wall with its base 5 ft from the wall. How high up the wall does it reach?



9. Find the length of the diagonal of a rectangle that is 9 in by 12 in.



10. Find the length of the diagonal of a square with side 7. Give the exact answer and a decimal to the nearest tenth.



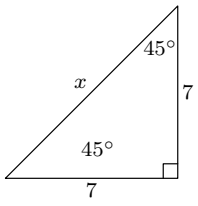
Geometry Review: Unit 5 — Right Triangles & Trigonometry

Station 2 of 4: Special Right Triangles: 45°-45°-90° and 30°-60°-90°

Name: _____ Date: _____

Give exact answers in simplest radical form (no decimals). Rationalize denominators. Diagrams are not drawn to scale.

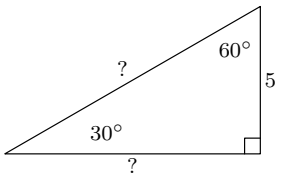
1. In a 45°-45°-90° triangle, each leg is 7. Find the hypotenuse.



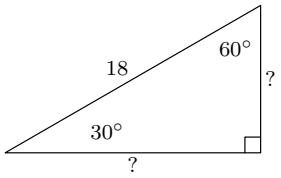
2. In a 45°-45°-90° triangle, the hypotenuse is 12. Find each leg.

3. In a 45°-45°-90° triangle, the hypotenuse is 10. Find each leg.

4. In a 30°-60°-90° triangle, the shorter leg is 5. Find the longer leg and the hypotenuse.



5. In a 30°-60°-90° triangle, the hypotenuse is 18. Find both legs.

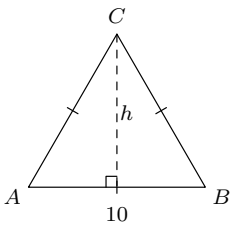


6. In a 30°-60°-90° triangle, the longer leg is $6\sqrt{3}$. Find the shorter leg and the hypotenuse.

7. In a 30°-60°-90° triangle, the longer leg is 9. Find the shorter leg and the hypotenuse.

8. The diagonal of a square is $8\sqrt{2}$. Find the side length.

9. Find the height of an equilateral triangle with side length 10.



10. Find the area of an equilateral triangle with side length 8.

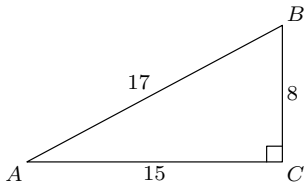
Geometry Review: Unit 5 — Right Triangles & Trigonometry

Station 3 of 4: Trigonometric Ratios: Sine, Cosine, Tangent

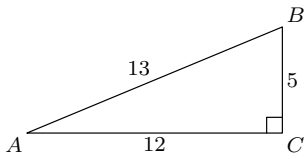
Name: _____ Date: _____

Write ratios as simplified fractions. Remember SOH-CAH-TOA. Diagrams are not drawn to scale.

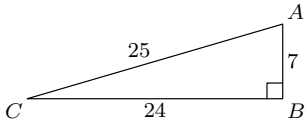
1. A right triangle has legs 8 and 15 and hypotenuse 17. $\angle A$ is opposite the leg of length 8. Find $\sin A$, $\cos A$, $\tan A$.



2. A right triangle has legs 5 and 12 and hypotenuse 13. $\angle B$ is opposite the leg of length 12. Find $\sin B$, $\cos B$, $\tan B$.



3. A right triangle has legs 7 and 24 and hypotenuse 25. $\angle C$ is opposite the leg of length 7. Find $\sin C$, $\cos C$, $\tan C$.



4. θ is acute and $\sin\theta = \dfrac{5}{13}$. Find $\cos\theta$ and $\tan\theta$.

5. θ is acute and $\cos\theta = \dfrac{3}{5}$. Find $\sin\theta$ and $\tan\theta$.

6. Complete: $\sin 20^\circ = \cos$ _____

7. In a right triangle, the side opposite θ is 6 and the side adjacent to θ is 8. Find $\tan\theta$, the hypotenuse, and $\sin\theta$.

8. Give the exact values: $\sin 30^\circ$, $\cos 60^\circ$, $\tan 45^\circ$, $\tan 60^\circ$.

9. In a right triangle, $\sin A = \dfrac{4}{5}$ and the side opposite $\angle A$ is 12. Find the hypotenuse and the adjacent side.

10. For the 3-4-5 triangle with $\angle A$ opposite 3, show that $\sin^2 A + \cos^2 A = 1$.

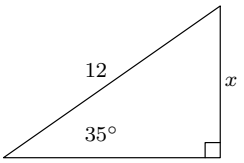
Geometry Review: Unit 5 — Right Triangles & Trigonometry

Station 4 of 4: Solving Right Triangles with Trigonometry

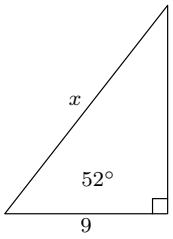
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Use a calculator in degree mode. Round lengths to the nearest tenth and angles to the nearest degree unless told otherwise. Diagrams are not drawn to scale.

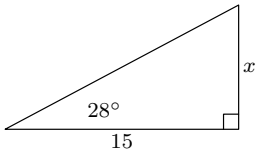
1. A right triangle has an angle of 35° and a hypotenuse of 12. Find the side x opposite the 35° angle.



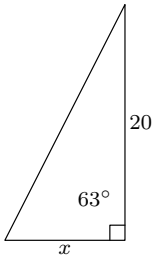
2. A right triangle has an angle of 52° and an adjacent side of 9. Find the hypotenuse x.



3. A right triangle has an angle of 28° and an adjacent side of 15. Find the opposite side x.



4. A right triangle has an angle of 63° and an opposite side of 20. Find the adjacent side x.

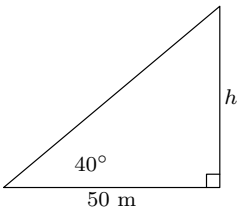


5. In a right triangle, the opposite side is 7 and the hypotenuse is 10. Find the angle θ to the nearest tenth of a degree.

6. In a right triangle, the adjacent side is 5 and the hypotenuse is 13. Find the angle θ to the nearest tenth of a degree.

7. In a right triangle, the opposite side is 9 and the adjacent side is 4. Find the angle θ to the nearest tenth of a degree.

8. From a point 50 m from the base of a tower, the angle of elevation to the top is 40°. How tall is the tower?



9. A 20-ft ladder makes a 70° angle with the ground. How high up the wall does it reach?

10. From the top of a 120-ft cliff, a boat is 200 ft from the base of the cliff. Find the angle of depression to the boat, to the nearest tenth of a degree.

